

Figure 4: Plotting multiple lines with different styles

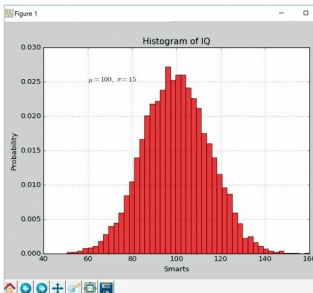


Figure 5: Histogram with text and grid

To plot X versus Y, we can use the commands shown below, as `plot()` takes an arbitrary number of arguments.

```
import matplotlib.pyplot as plt
plt.plot([1, 2, 3, 4], [1, 4, 9, 16])
plt.xlabel('X values')
plt.ylabel('Y values')
plt.show()
```

The output of the code above is shown in Figure 3.

To control the axis, `matplotlib` has the `axis()` command; it takes a list of values (Xmin, Xmax, Ymin, Ymax) and also specifies the viewport of the axes.

To plot several lines with different styles, use the following commands:

```
import matplotlib.pyplot as plt
import numpy as np
x = np.arange(10)
plt.plot(x, x, 'g')
plt.plot(x, 2 * x, 'r--')
plt.plot(x, 3 * x, 'g^')
plt.plot(x, 4 * x, 'bs')
plt.legend(['y = x', 'y = 2x', 'y = 3x', 'y = 4x'],
loc='upper left')
plt.show()
```

Controlling line properties

We can set many attributes to a line. The Table 1 shows a list of available *Line2D* properties.

Adding text into the plot

The `text()` command can be used to add text in an arbitrary location. The `xlabel()`, `ylabel()` and `title()` are used to add text in the indicated locations. The code below creates a histogram and some text. The output is shown in Figure 5.

```
import numpy as np
import matplotlib.pyplot as plt
mu, sigma = 100, 15
x = mu + sigma * np.random.randn(10000)
# the histogram of the data
n, bins, patches = plt.hist(x, 50, normed=1, facecolor='g',
alpha=0.75)
plt.xlabel('Smarts')
plt.ylabel('Probability')
plt.title('Histogram of IQ')
plt.text(60, .025, r'$\mu=100,\ \sigma=15$')
plt.axis([40, 160, 0, 0.03])
plt.grid(True)
plt.show()
```

This is only an introductory article on `matplotlib`. Next month, we will explore the topic in greater depth along with examples. The code is taken from <http://matplotlib.org> and readers are advised to look at the examples provided at <http://matplotlib.org/examples/index.html>. 

References

[1] <http://matplotlib.org/examples/index.html>

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